

```
1. #include <stdio.h>
```

```
#include <signal.h>
```

```
#include <unistd.h>
```

```
void handler(int sig)
```

```
{
```

```
printf("\nSIGINT received!\n");
```

```
printf("Signal number = %d\n", sig);
```

```
}
```

```
int main()
```

```
{
```

```
signal(SIGINT, handler);
```

```
while (1)
```

```
{
```

```
printf("Program is running...\n");
```

```
sleep(2);
```

```
}
```

```
return 0;
```

```
}
```

```

saranya@saisaranya:~$ gcc signal.c -o signal
saranya@saisaranya:~$ ./signal
Program is running...
Program is running...
Program is running...
^C
SIGINT received!
Signal number = 2
Program is running...
^C
SIGINT received!
Signal number = 2
Program is running...
Program is running...
^\\Quit
saranya@saisaranya:~$ █

```

2. #include <stdio.h>

#include <signal.h>

#include <unistd.h>

void handler(int sig)

{

if (sig == SIGTERM)

printf("\nSIGTERM received!\n");

}

int main()

{

signal(SIGTERM, handler);

printf("Process ID: %d\n", getpid());

printf("Waiting for SIGTERM...\n");

while (1)

{

sleep(2);

```
}
```

```
return 0;
```

```
}
```

```
saranya@saisaranya:~$ gcc sigterm.c -o sigterm
saranya@saisaranya:~$ ./sigterm &
[1] 122
saranya@saisaranya:~$ Process ID: 122
Waiting for SIGTERM...
kill -SIGTERM 122
saranya@saisaranya:~$
SIGTERM received!
kill -9 122
saranya@saisaranya:~$
[1]+  Killed                  ./sigterm
saranya@saisaranya:~$
```

```
3. #include <stdio.h>
```

```
#include <signal.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
void child_handler(int sig)
```

```
{
```

```
printf("SIGCHLD received: Child has terminated\n");
```

```
wait(NULL);
```

```
}
```

```
int main()
```

```
{
```

```
signal(SIGCHLD, child_handler);
```

```
pid_t pid = fork();
```

```
if (pid == 0)
```

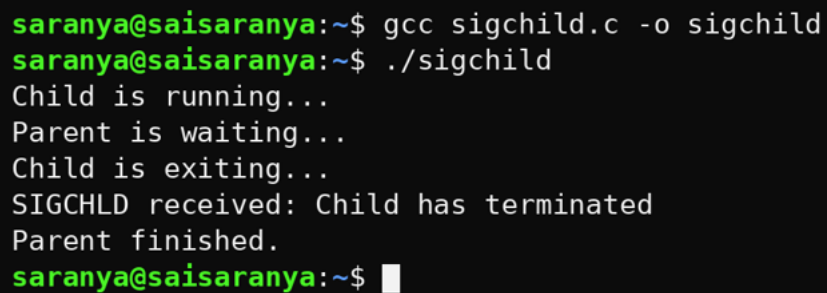
```
{
```

```

printf("Child is running...\n");
sleep(2);
printf("Child is exiting...\n");
}
else
{
printf("Parent is waiting...\n");
sleep(5);
printf("Parent finished.\n");
}

return 0;
}

```



```

saranya@saisaranya:~$ gcc sigchild.c -o sigchild
saranya@saisaranya:~$ ./sigchild
Child is running...
Parent is waiting...
Child is exiting...
SIGCHLD received: Child has terminated
Parent finished.
saranya@saisaranya:~$

```

4. #include <stdio.h>

#include <unistd.h>

#include <signal.h>

#include <stdlib.h>

void handle_sigpipe(int sig)

```

{
printf("\nSIGPIPE received!\n");
printf("The reading end of the pipe is closed.\n");
exit(1);
}

```

```

int main()
{
int fd[2];

signal(SIGPIPE, handle_sigpipe);

pipe(fd);

printf("Pipe created.\n");

// Close the reading end
close(fd[0]);

printf("Reading end closed.\n");

printf("Trying to write to the pipe...\n");

write(fd[1], "Hello", 5);

printf("Write completed.\n");

return 0;
}

```

```

saranya@saisaranya:~$ gcc sigpipe.c -o sigpipe
saranya@saisaranya:~$ ./sigpipe
Pipe created.
Reading end closed.
Trying to write to the pipe...

SIGPIPE received!
The reading end of the pipe is closed.
saranya@saisaranya:~$ echo $?
1
saranya@saisaranya:~$ █

```

5. Program

```
#include <stdio.h>
```

```
#include <signal.h>
#include <stdlib.h>

void handle_sigsegv(int sig)
{
    printf("\nSIGSEGV received!\n");
    printf("Invalid memory access detected.\n");
    exit(1);
}

int main()
{
    signal(SIGSEGV, handle_sigsegv);

    printf("Program started.\n");

    int *p = NULL;

    printf("Trying to access invalid memory...\n");

    *p = 10;

    printf("Program completed.\n");

    return 0;
}
```

```
saranya@saisaranya:~$ gcc sigsegv.c -o sigsegv
saranya@saisaranya:~$ ./sigsegv
Program started.
Trying to access invalid memory...

SIGSEGV received!
Invalid memory access detected.
saranya@saisaranya:~$ echo $?
1
saranya@saisaranya:~$ █
```